

Cheng Huang

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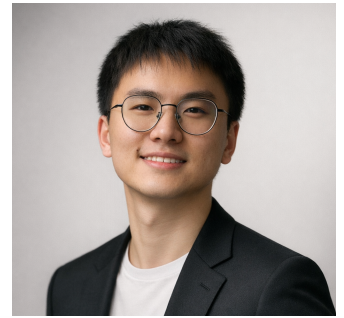
Email: huangchengadam@gmail.com

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GitHub: https://github.com/kouteisang

Location: Based on Copenhagen, Denmark | Open to relocation

Personal Website: Cheng Huang's Personal Website



WORK EXPERIENCE

FOSS

Software Engineer – Algorithm Implementation | R&D Machine Learning Team

Hillerød, Denmark

Jul. 2026 – Present

- **Algorithm Engineering:** Implement and optimize algorithms for production machine learning and imaging systems, translating Python prototypes into efficient C# implementations.
- **GPU Computing:** Design, implement, and optimize CUDA kernels to accelerate performance-critical algorithms in production software.
- **Software Engineering & Collaboration:** Develop and maintain production software components, collaborating with ML researchers and software engineers on implementation, testing, code reviews, and integration.

Aarhus University

Ph.D. in Computer Science, Advisors: Prof. Davide Mottin, Prof. Ira Assent

Aarhus, Denmark

Jun. 2023 – Jun. 2026

- **Research:** Designed and implemented high-performance parallel graph algorithms on multi-core CPUs and NVIDIA GPUs, with large-scale experimental evaluation on real-world graphs.
- **Teaching & Supervision:** Database, Machine Learning, Deep Learning for Visual Recognition; supervised Bachelor's theses.

ByteDance (TikTok)

Software Engineer Intern

Beijing, China

Feb. 2021 – Jun. 2021

- **Test Automation:** Developed Python-based automated testing scripts for web applications, integrating database operations and data validation to support functional and regression testing.
- **Debugging & Reliability:** Investigated test failures, analyzed application and database behavior, and collaborated with engineers to identify root causes and improve software reliability.

EDUCATION

Aarhus University

Ph.D. in Computer Science

Aarhus, Denmark

Jun. 2023 – Jun. 2026

- **Thesis:** Scalable Graph Analytics on Modern Parallel Architectures

Aarhus University

M.Sc. Computer Science, GPA: 10.9/12, Danish State Scholarship with 30,000 Euro

Aarhus, Denmark

Sep. 2021 – Jul. 2023

Shandong University of Technology

B.Eng. Software Engineering, GPA: 90.1/100

Shandong, China

Sep. 2017 – Jul. 2021

SELECTED RESEARCH PROJECTS

D-Core | C++, CUDA, OpenMP, Python, Git; Parallel Graph Algorithms on CPU/GPU

- Designed parallel and scalable D-Core decomposition algorithms for billion-scale real-world directed graphs.
- Implemented GPU algorithms that address thread-level workload imbalance and reduce algorithmic memory overhead, removing key performance bottlenecks in practice.
- Evaluated on billion-scale real-world graphs, achieving up to 47× speedup over the previous best method while reducing memory usage by 33%.

FirmCore | C++, CUDA, OpenMP, Python, Git; Parallel Graph Algorithms on CPU/GPU

- Designed resource-efficient and scalable parallel FirmCore decomposition algorithms for billion-scale real-world multilayer graphs.
- Implemented parallel algorithms that maximize parallel resource utilization by reducing idle threads on multi-core CPUs, and proposed hierarchical parallelism to further expose parallelism on GPUs.
- Evaluated on billion-scale real-world graphs, achieving up to 100.3× speedup over the previous best method.

SKILLS & OTHERS

Programming Languages: C#, CUDA, C++, Python, Java, SQL

Technical Expertise: Parallel Algorithms, GPU Computing, High-Performance Computing, Performance Optimization, Large-Scale Data Analytics, Data Mining, Machine Learning

Frameworks & Tools: OpenMP, PyTorch, TensorFlow, Git, Linux, CMake, NVIDIA Nsight Systems

Volunteer: Supported children with intellectual disabilities during undergraduate studies.

PUBLICATIONS & MANUSCRIPTS

- **Geld: Load-balanced D-Core Decomposition for Consumer GPUs**
Cheng Huang, Johannes Langguth, Xing Cai, Davide Mottin, Ira Assent
ACM International Conference on Management of Data (SIGMOD 2026) (CORE Ranking: A*)
- **Resource-Efficient FirmCore Decomposition on Billion-scale Multilayer Graphs**
Cheng Huang, Davide Mottin, Ira Assent
International Conference on Very Large Data Bases (VLDB 2026) (CORE Ranking: A*)
- **HunIPU: Efficient Hungarian Algorithm on IPU**s
Cheng Huang, Alexander Mathiasen, Josef Dean, Davide Mottin, Ira Assent
The 40th IEEE International Conference on Data Engineering (ICDE) Workshops: HardDB & Active
- **GCore: A Fast GPU-parallelized Approach to D-Core Decomposition**
Cheng Huang, Johannes Langguth, Davide Mottin, Ira Assent
9th Joint Workshop on Graph Data Management Experiences & Systems (GRADES) and Network Data Analytics (NDA)
- **Accelerating the Hungarian Algorithm: Computing Linear Assignments on CPUs, GPUs, and IPU**s
Cheng Huang, Davide Mottin, Ira Assent, Johannes Langguth
ACM Transactions on Mathematical Software (Under minor revision)